

Research Paper

Numerical study on temperature distribution uniformity and cooling performance of manifold microchannel heat sink

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ABSTRACT

The manifold microchannel heat sink (MMCHS) is a promising design for cooling electronic devices due to its ability to effectively dissipate high heat flux. The structural changes of the heat sink play a crucial role in achieving uniform heat dissipation and optimal cooling performance. In order to enhance temperature distribution uniformity and cooling performance, this study conducted numerical simulations by introducing flow-regulating baffles between the manifold channels and microchannels. Additionally, two different types of diverging/converging manifold channels, namely horizontal (z direction) and vertical (y direction) diverging/converging manifold channels, were examined. The simulations were performed under single-phase conditions with varying coolant mass flow rates, and various parameters such as flow distribution, pressure drop, thermal resistance, temperature uniformity, and overall heat transfer performance were analyzed. The findings indicate that the modified MMCHS with baffles exhibited improved temperature distribution uniformity at the base of the heat sink and reduced average thermal resistance. However, it was observed that the pressure drop increased in all cases. The MMCHS with baffles demonstrated an average increase of 70.68% in microchannel flow distribution uniformity, an average increase of 44.55% in temperature distribution uniformity at the base of the heat sink, and an average increase of 7.67% in cooling coefficient of performance (COP). On the other hand, the MMCHS with diverging/converging manifold channels in the z and y directions showed an average increase of 81.44% and 83.65% in microchannel flow distribution uniformity, and an average increase of 57.71% and 62.13% in temperature distribution uniformity. However, it was noted that the COP decreased by an average of 14.58% and 29.81% for these cases.

1. Introduction

In recent decades, the continuous integration and miniaturization of electronic devices have led to an increase in heat flux. Excessively high temperatures can negatively impact the performance and reliability of electronic devices, making it necessary to meet strict heat dissipation requirements. The micro- and power-electronics sectors face the challenge of dissipating high heat flux while keeping the temperature below 85°C to prevent overheating [1]. Thermal problems contribute to more than 50 % of integrated circuit failures [2]. Research has shown that a decrease of 10 °C in junction temperature can reduce the failure rate of electrical equipment by half [3]. Hence, it is crucial to develop efficient heat dissipation devices for electronic components.

In 1981, Tuckerman and Pease [4] first proposed the use of microchannel heat sinks (MCHS) for liquid cooling of chips. These heat sinks were able to dissipate a heat flux of about 790 W/cm² in the heat-

generating area of the chip. The findings showed that the highest temperature rise was 71 °C higher than the coolant inlet temperature. As a result, MCHSs became one of the most effective methods for chip heat dissipation. However, traditional microchannel heat sinks (TMCHS) have two downsides. Firstly, the flow of coolant in long straight microchannels for heat exchange leads to poor temperature distribution uniformity [5]. Secondly, there is excessive pressure drop [6]. To address these shortcomings, Harpole and Enginger [7] proposed a manifold microchannel heat sink in 1991 (see Fig. 1). This design introduced a manifold shunt system above the microchannels and divided the microchannel into many parallel U-shaped channels, shortening the flow path. This increased fluid disturbance in the microchannels, reduced pressure drop and thermal resistance, and improved temperature distribution uniformity. Subsequent studies have confirmed that the manifold microchannel heat sink (MMCHS) exhibits better thermal performance compared to TMCHS [8,9]. Various investigations have also been conducted to further enhance the

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Nomenclature		δ	mean absolute temperature deviation, K
Q	heat absorbed by coolant, W	<i>Subscripts</i>	
m	mass flow rate, kg/s	<i>eff</i>	effective
c_p	specific heat capacity, J/(kg·K)	<i>in</i>	inlet
T	temperature, K	<i>out</i>	outlet
A	area, m ²	<i>tot</i>	total
q	heat flux, W/m ²	<i>heat</i>	coolant absorbs heat
R	thermal resistance, K/W	<i>conv</i>	convection
COP	cooling coefficient of performance	<i>cond</i>	conduction
Re	Reynolds number	<i>ref</i>	reference
p	pressure, kPa	<i>f</i>	fluid/fin
P	Pumping power, W	<i>r</i>	relative
Si	Silicon	<i>i</i>	i-th microchannel
v	volume flow rate, m ³ /s	<i>hs</i>	heat sink
l	length, μ m	<i>base</i>	heat sink base
h	height, μ m/ convective heat transfer coefficient, W/(m ² ·K)	<i>max</i>	maximum value
w	width, μ m	<i>min</i>	minimum value
Nu	Nusselt number	T	temperature
D_h	characteristic size, m	<i>div</i>	divider
<i>Greek symbols</i>		<i>b</i>	baffle
ρ	density, kg/m ³	<i>c</i>	microchannel
μ	dynamic viscosity, kg/(m·s)	<i>m</i>	manifold channel
Δ	difference	<i>sur</i>	surface
k	thermal conductivity, W/(m·K)	<i>gl</i>	global
s^2	variance	<i>lo</i>	local

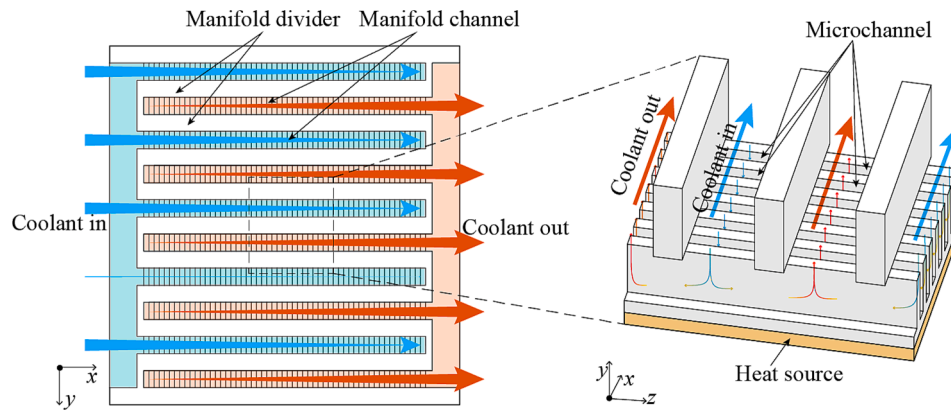


Fig. 1. Schematic diagrams of typical MMCHS.

performance of MMCHS.

The existing research has explored the size parameters and structure of microchannels, the size parameters and structure of manifold channels, the study of different working fluids and heat sink materials, the study of flow boiling heat transfer, and so on. Suchismita et al. [10] examined the impact of geometric factors on MMCHS performance, including microchannel depth and width, manifold channel depth, and manifold channel inlet and outlet lengths. Their findings revealed that increasing the aspect ratio of microchannels leads to higher Reynolds (Re) and Nusselt numbers (Nu). However, the depth of the manifold channel does not significantly affect the heat sink heat transfer coefficient. Additionally, they found that the best heat transmission occurs when the ratio of manifold channel inlet to outlet length is three. In another study, Kim et al. [11] investigated the effect of microchannels with different widths and heights, using air as the working medium. They concluded that the thermal resistance of MMCHS is lower than that

of TMCHS under the same operating conditions. Furthermore, they observed that the thermal resistance is primarily influenced by the flow rate, microchannel width, and manifold inlet and outlet width. Escher et al. [12] conducted a study on the impact of the number and width of microchannels, as well as the flow rate of MMCHS, on heat transfer performance and pump power. They determined that the optimal arrangement of a manifold type of microchannel heat sink depends on the width of the manifold channel and the width of the microchannels. They also provided the optimal size design for the given operating conditions. Andhare et al. [13] designed a flat plate MMCHS and tested it using water as the working fluid in single-phase. They achieved a total heat transfer coefficient of approximately 20,000 W/m²K at flow rates as low as 20 g/s, with a pressure drop per unit length of 5.85 bar/m. Van Erp et al. [8] evaluated the influence of the number of inlet and outlet manifold channels on the heat transfer performance of the MMCHS using water as the working fluid and silicon as the material. They found

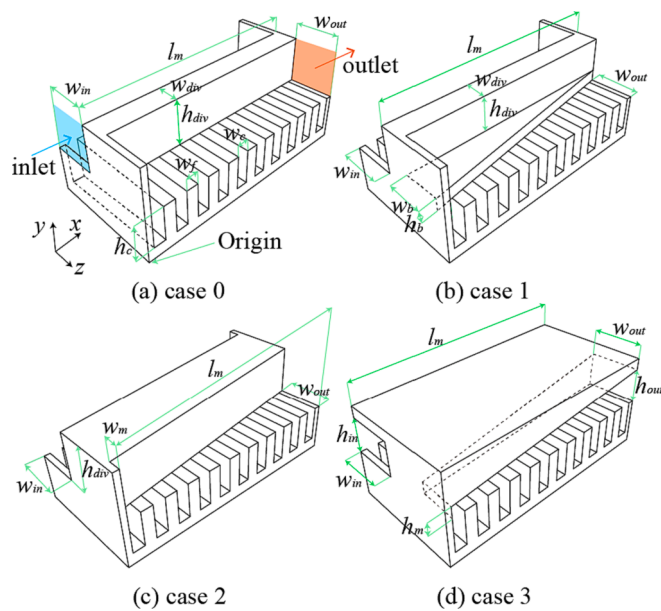


Fig. 2. Schematic diagram of the model.

that increasing the number of inlet and outlet manifold channels improved the heat transfer performance of the MMCHS significantly, resulting in a COP value 50 times greater than that of TMCHS. Yang et al. [14] proposed a novel hybrid microchannel heat sink that combines a manifold with secondary oblique channels. They discovered that this design effectively reduces both thermal resistance and pressure drop, outperforming the MMCHS. Xie et al. [15] conducted a study on the impact of different coolants and microchannel depths on heat transfer and flow in the MMCHS. Their findings revealed that water exhibited superior performance compared to HFE-7100. Furthermore, increasing the microchannel depth resulted in reduced pressure drop while maintaining the same flow rate. Chen et al. [16,17] investigated the influence of porous metal fins and various manifold types on MMCHS flow and heat transfer properties. The inclusion of porous fins led to a 19 % decrease in pressure drop, improved flow distribution, and enhanced heat transmission in the MMCHS. Among the different manifold types, the parabolic manifold demonstrated the highest overall performance, while the conical manifold exhibited the lowest pressure drop. The elliptical manifold offered the best temperature uniformity. Lin et al. [18] conducted a numerical analysis on the single-phase and two-phase heat transfer and pressure drop characteristics of different manifold microchannel heat sinks, including Z-type, C-type, U-type, and H-type. Their study explored the fluid flow distribution and temperature uniformity within these MMCHSs. The results showed that different types of MMCHSs significantly influence flow distribution and temperature uniformity. Notably, the U-type MMCHS exhibited a good uniformity of temperature distribution.

Pan et al. [19,20] conducted a study on the effect of the micro-channel aspect ratio on the heat transfer performance of MMCHSs. They found that the optimal aspect ratio increases with the thermal conductivity of the channel wall and the dynamic viscosity of the working fluid. They developed a standard formula to describe the relationship between

each thermophysical parameter and the optimal aspect ratio. The researchers also designed and analyzed a cylindrical pin-fin MMCHS. Their findings revealed that, compared to the rectangular MMCHS, the pin-fin MMCHS exhibited better heat transfer performance but had a higher pressure drop. Moreover, the pin-fin MMCHS outperformed the rectangular MMCHS in terms of heat transfer performance at the same pressure drop. Mandel et al. [21] employed numerical simulations to investigate the impact of manifold channel type on the pressure drop of single-phase flow and flow boiling in MMCHS. They observed that Z-type manifold channels provided better flow distribution at low flow rates, C-type manifold channels offered better flow distribution at high flow rates, and H-type manifold channels exhibited a more even pressure distribution in flow boiling conditions. Tang et al. [22] proposed a new MMCHS with diverging/converging microchannels and conducted numerical investigations on its flow and heat transfer capabilities under single-phase conditions. They explored the effects of the manifold inlet-to-outlet ratio α and the channel inlet-to-outlet ratio β . The results indicated that moderately divergent manifold layouts and converging channel layouts yielded the best heat transfer improvement for a given pump power. Additionally, several studies have examined the influence of introducing nanofluids and boiling heat transfer [18,23–32].

Previous studies have shown that MMCHS exhibits better heat transfer capability and temperature distribution compared to TMCHS. However, the presence of an obstruction at the end of the inlet manifold channel leads to a significant amount of fluid being forced into adjacent microchannels, resulting in uneven flow and temperature distribution. This non-uniform heat dissipation can lead to localized high temperatures and uneven thermal stress, which can potentially impact the

Table 2
The physical properties of water and Si.

Parameters	Water	Si
$\rho(\text{kg}/\text{m}^3)$	983.2	2300
$\mu(\text{kg}/(\text{m}\cdot\text{s}))$	0.001003	*
$c_p(\text{J}/\text{kg})$	4185	700
$k(\text{W}/\text{m}\cdot\text{K})$	0.651	150

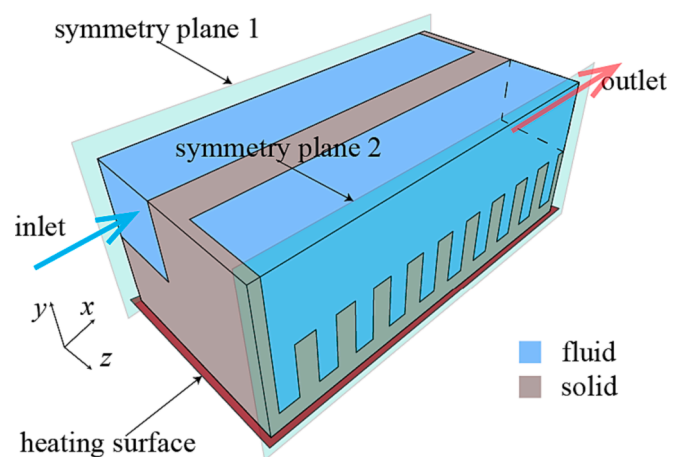


Fig. 3. Schematic of the boundary conditions.

Table 1
Summary of the dimensions of MMCHS, unit: μm .

Model	w_{in}	w_{out}	w_{div}	w_m	l_m	h_{div}	w_b	h_b	h_{in}	h_{out}	h_m	h_c	w_c	w_f
case 0	200	200	100	*	5000	250	*	*	*	*	*	200	50	50
case 1	200	200	100	*	5000	200	150	50	*	*	*	200	50	50
case 2	200	50	*	50	5000	250	*	*	*	*	*	200	50	50
case 3	200	200	100	*	5000	*	*	*	250	250	25	200	50	50

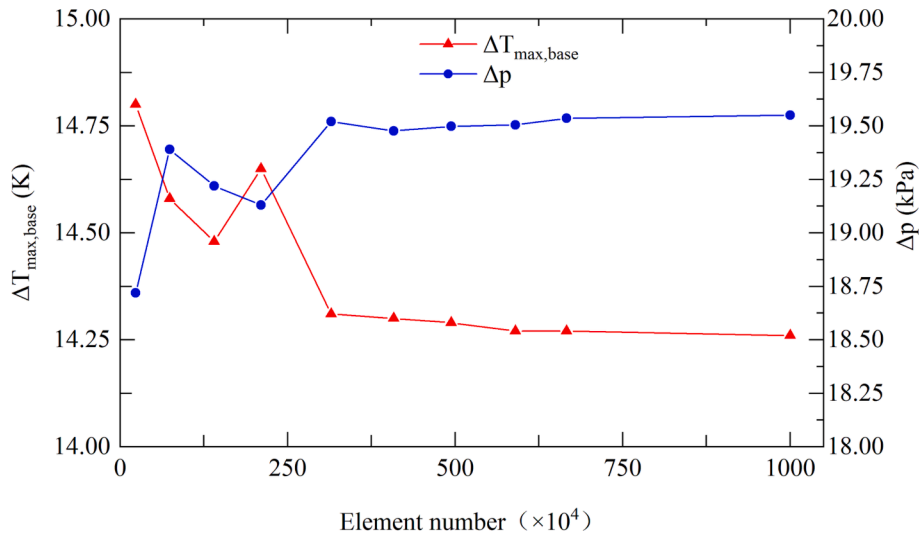


Fig. 4. Pressure drop and temperature at different grid numbers.

Table 3
Grid dependence validation and quality.

Model	Element number	$\Delta T_{max,base}$ (K)	E (%)	Δp (Kpa)	E (%)	Maximum skewness	Minimum orthogonality quality	Select
case 1	1496361	9.52	1.35	28.14	−0.07	0.17	0.83	✓
	3129252	9.46	1.96	28.19	0.11	0.15	0.85	
	5367858	9.57	0.83	28.20	0.14	0.15	0.85	
	7318493	9.64	0.17	28.18	0.07	0.17	0.83	
	12498797	9.65	*	28.16	*	0.15	0.85	
case 2	1431003	7.88	3.43	38.66	0.21	0.16	0.84	✓
	3423615	8.09	0.86	38.51	−0.18	0.17	0.83	
	5233834	8.08	0.98	38.60	0.05	0.15	0.85	
	7237430	8.14	0.24	38.56	−0.05	0.19	0.81	
	12329347	8.16	*	38.58	*	0.15	0.85	
case 3	1675593	8.40	−3.78	46.44	1.35	0.17	0.83	✓
	3870567	8.54	−2.17	46.22	0.87	0.15	0.85	
	5622649	8.59	−1.60	45.98	0.35	0.17	0.83	
	7484210	8.66	−0.80	45.92	0.22	0.16	0.84	
	13402019	8.73	*	45.82	*	0.16	0.84	

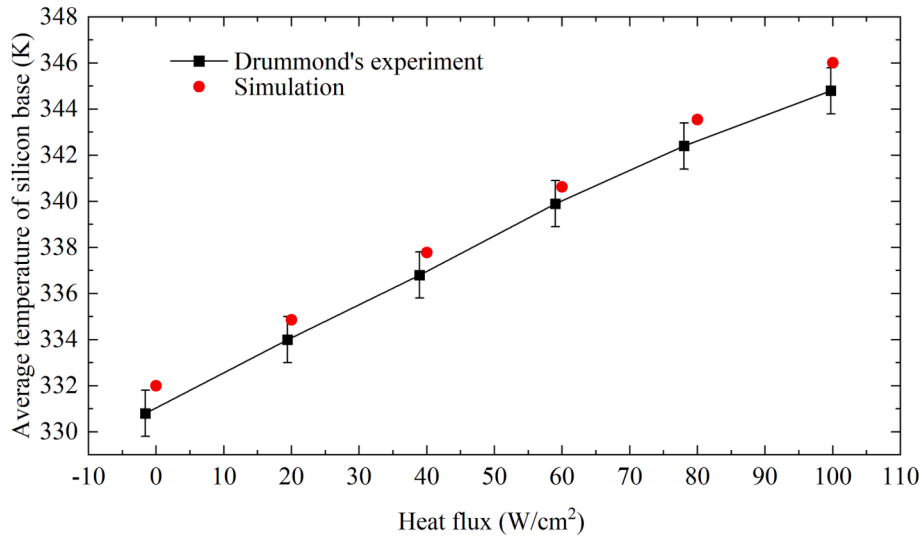


Fig. 5. Comparison of experimental and numerical simulations.

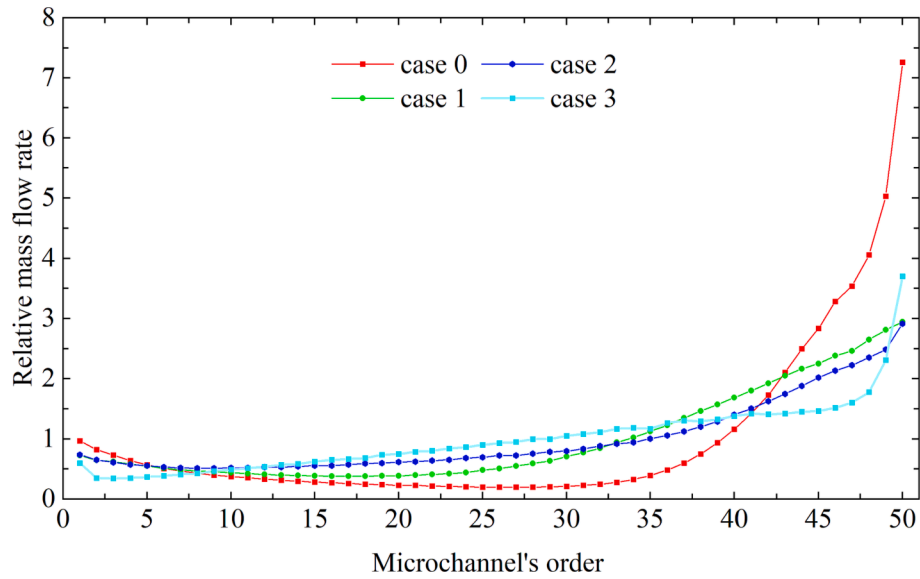


Fig. 6. Flow distribution diagram of microchannels.

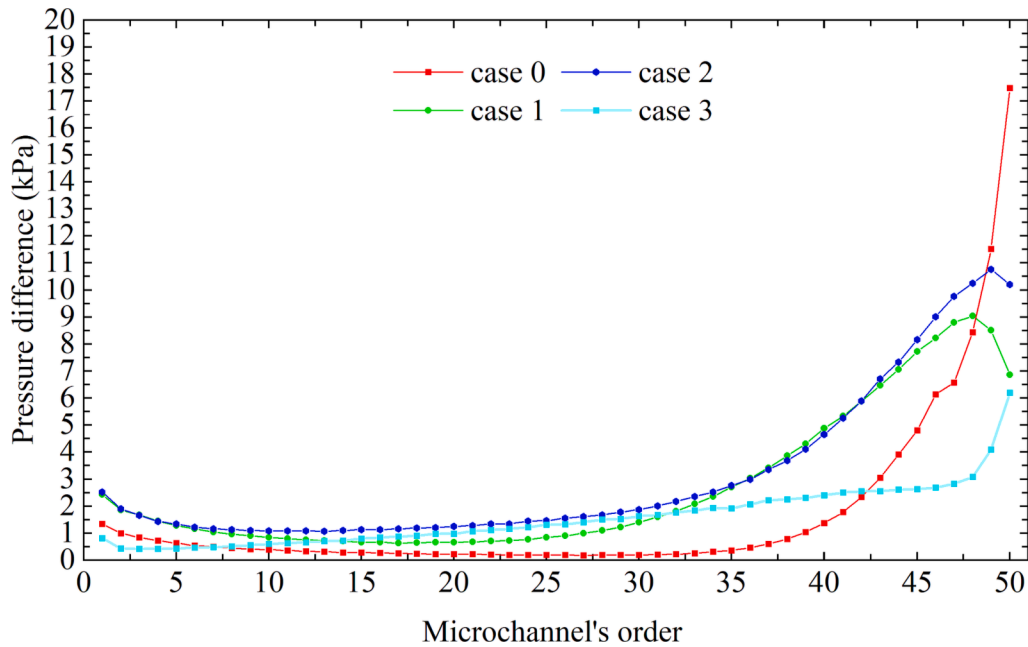


Fig. 7. The static pressure difference between the inlet and outlet of the microchannel.

performance, stability, and service life of electronic equipment. Some researchers have explored the use of U-shaped MMCHS or reverse flow arrangement of manifold channels and microchannels, and have observed improvements in flow distribution and temperature uniformity to some extent. However, the practical application of these approaches in engineering may be limited due to the large size of the heat sink or the complexity of channel arrangement. This study innovatively introduces baffles between manifold channels and microchannels to improve flow distribution within microchannels. It also expands the research on divergent/convergent manifold channels. Numerical simulation methods were used to study the flow and heat transfer characteristics of heat sinks under different inlet flow rates under single-phase conditions. Previous studies mainly focused on improving comprehensive heat transfer performance, with relatively little analysis and discussion on the uniformity of temperature distribution. Therefore, this study separately analyzed and discussed the uniformity of temperature distribution as an

indicator, and evaluated the comprehensive heat transfer performance. The findings of this study will provide theoretical reference for designing microchannel heat sinks with simpler structure, better heat transfer performance, and more uniform temperature distribution.

2. Numerical simulation method

2.1. Physical model

The typical MMCHS schematic diagram in Fig. 1 shows that MMCHS consists of multiple parallel manifold channels and microchannels with the same width and height. The channels in the middle area of the heat sink exhibit periodic and symmetrical characteristics. To save computational resources, it is possible to focus on studying only one unit structure. Fig. 2 presents a schematic diagram of the unit structure under investigation, but it does not depict the actual number of microchannels,

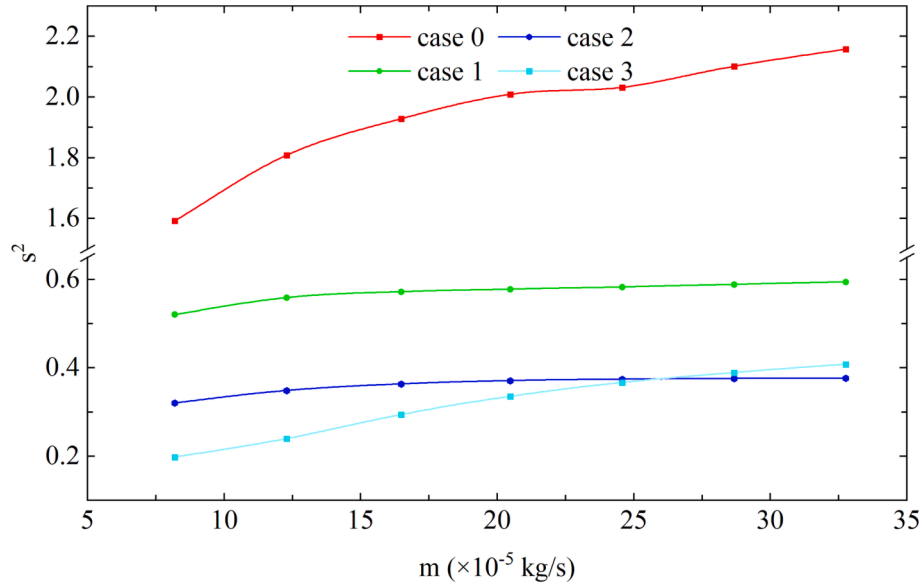


Fig. 8. Unevenness of flow distribution at different flow rates.

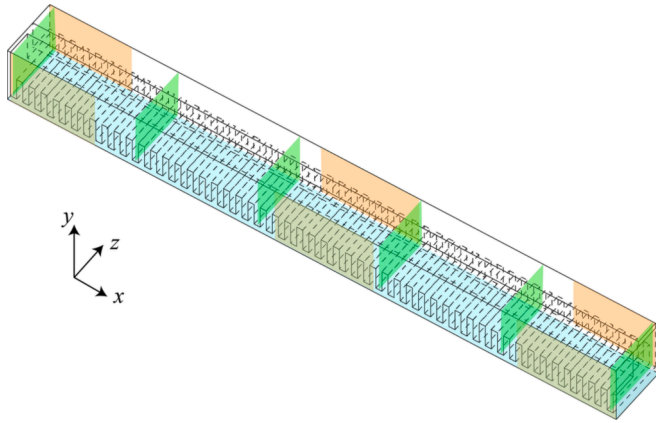


Fig. 9. Selection of some planes.

which is 50. Case 0 serves as a reference object for typical MMCHS research. In case 1, a baffle is added between the manifold channel and the microchannel to adjust the flow rate. Case 2 and case 3 utilize divergent/convergent manifold channels. The difference lies in the direction of divergence/convergence, with case 2 diverging/converging in the z direction and case 3 in the y direction. For specific dimensions, please refer to Table 1.

2.2. Governing equations and boundary conditions

The flow and heat transfer of MMCHS were simulated using the computational fluid dynamics program ANSYS Fluent 2022. A three-dimensional fluid–structure coupling model was employed for the simulation. The pressure–velocity coupling heat transfer problem was solved using finite volume techniques and a simple algorithm. The continuity equation was solved using the interpolation method of momentum-based averaging filter, while the gradient was discretized using the least squares cell-based method. The pressure, momentum, and energy equations were discretized using the second-order upwind method. Convergence of the calculation was determined by monitoring the residuals of the continuity equation, momentum equation, and energy equation, which were set to 10^{-6} and 10^{-9} respectively. The calculation was considered to have converged when the residual values

were difficult to further reduce and the curve of the average temperature at the base of the heat sink became almost horizontal.

The following are the major assumptions:

- (1) The fluid flow is steady, incompressible, single-phase, and laminar.
- (2) The physical properties of the fluid and solid are constant.
- (3) Gravity effects and heat losses to the environment are neglected.
- (4) The solid medium is homogeneous and isotropic.

Based on the above assumptions, the relevant control equations are as follows:

Continuity equation:

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

Momentum equations:

$$(\mathbf{u} \cdot \nabla) \rho_f \mathbf{u} = -\nabla P + \mu_f \nabla^2 \mathbf{u} \quad (2)$$

Energy equation for the fluid region:

$$\mathbf{u} \cdot \nabla T = \frac{k_f}{\rho_f c_{p,f}} \nabla^2 T \quad (3)$$

Energy equation for the heat sink solid region:

$$k_s \nabla^2 T = 0 \quad (4)$$

Where ρ_f and μ_f is the density and the dynamic viscosity of the coolant, k_f and $c_{p,f}$ are the thermal conductivity and specific heat of the coolant, and k_s is the thermal conductivity of the solid.

The coolant used in this study is water, and the solid component of the heat sink is made of silicon. The thermophysical properties of fluids and solids are presented in Table 2. The boundary condition is depicted in Fig. 3, with the inlet temperature set at 298 K. The flow rate ranges from $8.192 \times 10^{-5} \text{ kg/s}$ to $3.2768 \times 10^{-4} \text{ kg/s}$, and the maximum Re at the inlet is 2006. The outlet boundary condition is assumed to be a pressure boundary condition, with the outlet pressure set to zero. Symmetric boundary conditions are applied to the manifold channels. The heating surface at the base of the heat sink is subjected to a constant heat flux ($1 \times 10^6 \text{ W/m}^2$), while the other surfaces have adiabatic boundary conditions.

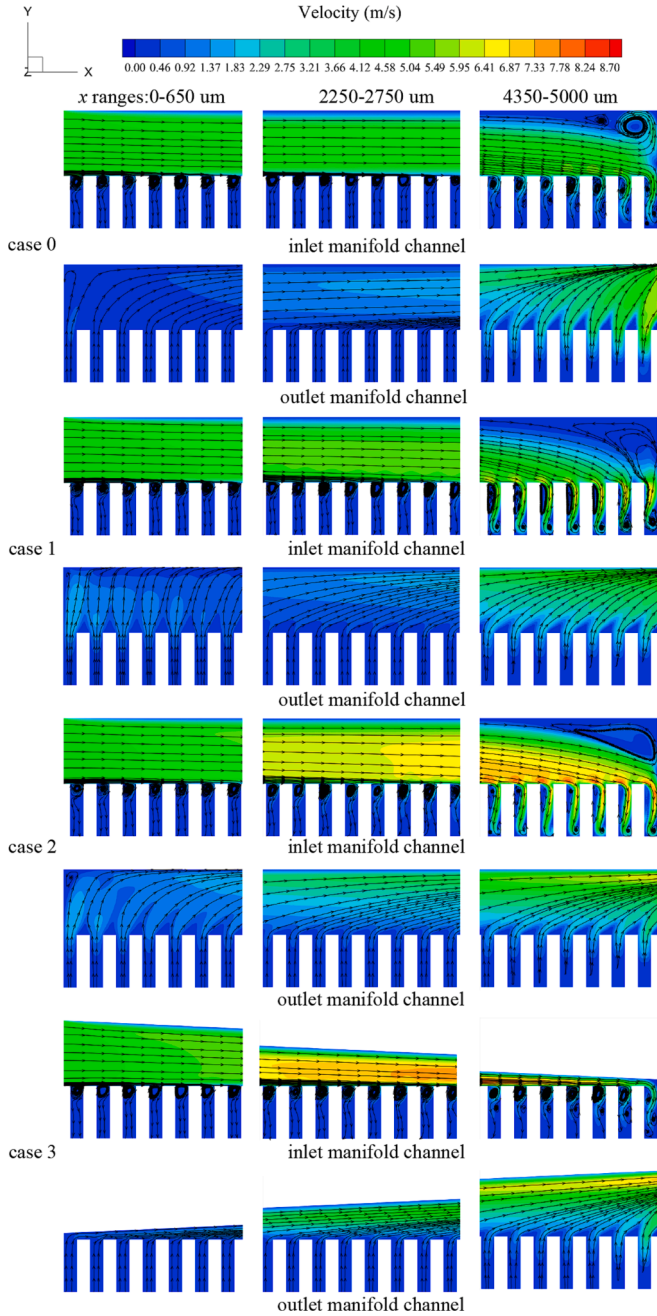


Fig. 10. Velocity distribution on the y-x planes.

2.3. Data reduction

2.3.1. Thermal resistance and Nusselt number

As the heat loss from the heat sink to the environment is ignored, all the heat is absorbed by the coolant, and the expression for the effective heat absorbed by the coolant is as follows:

$$Q_{eff} = mc_p \Delta T = q_{base} A_{base} \quad (5)$$

Where m is the mass flow rate, c_p is the specific heat of the coolant, q_{base} is the heat flux from the base of the heat sink, and ΔT is the difference in temperature between the coolant at the outlet of the heat sink and the coolant at the inlet. A_{base} is the heating area at the base of the heat sink.

Total thermal resistance (R_{tot}) is divided into three parts by the one-dimensional heat transfer model [33], including the thermal resistance caused by the finite heat capacity of water (R_{heat}), the thermal resistance

caused by convective heat transfer (R_{conv}), and the thermal resistance caused by thermal conduction (R_{cond}).

R_{heat} is correlates to the fluid's density, specific heat and flow rate. If the fluid's reference temperature is T_{ref} , then the fluid's average thermal resistance $R_{heat,mean}$ is as follows:

$$R_{heat,mean} = \frac{T_{ref} - T_{in}}{Q_{eff}} = \frac{1}{2} \frac{1}{\rho_f C_p v} \quad (6)$$

$$T_{ref} = \frac{T_{in} + T_{out}}{2} \quad (7)$$

$$v = \frac{m}{\rho_f} \quad (8)$$

Where ρ_f is the coolant density, v is the volume flow rate, T_{in} and T_{out} are the inlet and outlet temperatures of the coolant.

If $T_{base,mean}$ is the average temperature of the base of the heat sink, the average convective thermal resistance $R_{conv,mean}$ is as follows:

$$R_{conv,mean} = \frac{T_{base,mean} - T_{ref}}{Q_{eff}} \quad (9)$$

The average thermal conductivity thermal resistance $R_{cond,mean}$ is expressed as:

$$R_{cond,mean} = \frac{T_{sur,mean} - T_{base,mean}}{Q_{eff}} \quad (10)$$

Where $T_{sur,mean}$ is the average surface temperature inside the channel.

Total average thermal resistance:

$$R_{tot} = R_{cond,mean} + R_{conv,mean} + R_{heat,mean} \quad (11)$$

The Nusselt number is used to reflect the intensity of convective heat transfer, which is divided into global Nusselt number (Nu_{gl}) and local Nusselt number ($Nu_{lo,i}$), and its expression is as follows:

$$Nu_{gl} = \frac{h_{gl} D_h}{k_f} \quad (12)$$

$$Nu_{lo,i} = \frac{h_i D_h}{k_f} \quad (13)$$

$$h_{gl} = \frac{q_{sur,mean}}{T_{sur,mean} - T_{ref}} \quad (14)$$

$$D_h = \frac{2w_c h_c}{(w_c + h_c)} \quad (15)$$

$$h_i = \frac{q_{sur,i}}{T_{sur,i} - T_{ave,i}} \quad (16)$$

Where h_{gl} is the global convective heat transfer coefficient, D_h is the channel characteristic size, $q_{sur,mean}$ is the average heat flux within the channel. The h_i , $q_{sur,i}$, $T_{sur,i}$ and $T_{ave,i}$ are the average convective heat transfer coefficient, average heat flux, average surface temperature, and average fluid temperature within the i -th microchannel.

2.3.2. Flow rate deviation and temperature deviation

Flow rate variance is used to evaluate the unevenness of flow distribution in microchannels, defined as follows:

$$s^2 = \frac{\sum_{i=1}^n (m_{r,i} - 1)^2}{n} \quad (17)$$

$$m_{r,i} = \frac{nm_i}{\sum_{i=1}^n m_i} \quad (18)$$

Where $m_{r,i}$ is the relative mass flow rate of the i -th microchannel and m_i is the mass flow rate of the i -th microchannel, n is the total number of

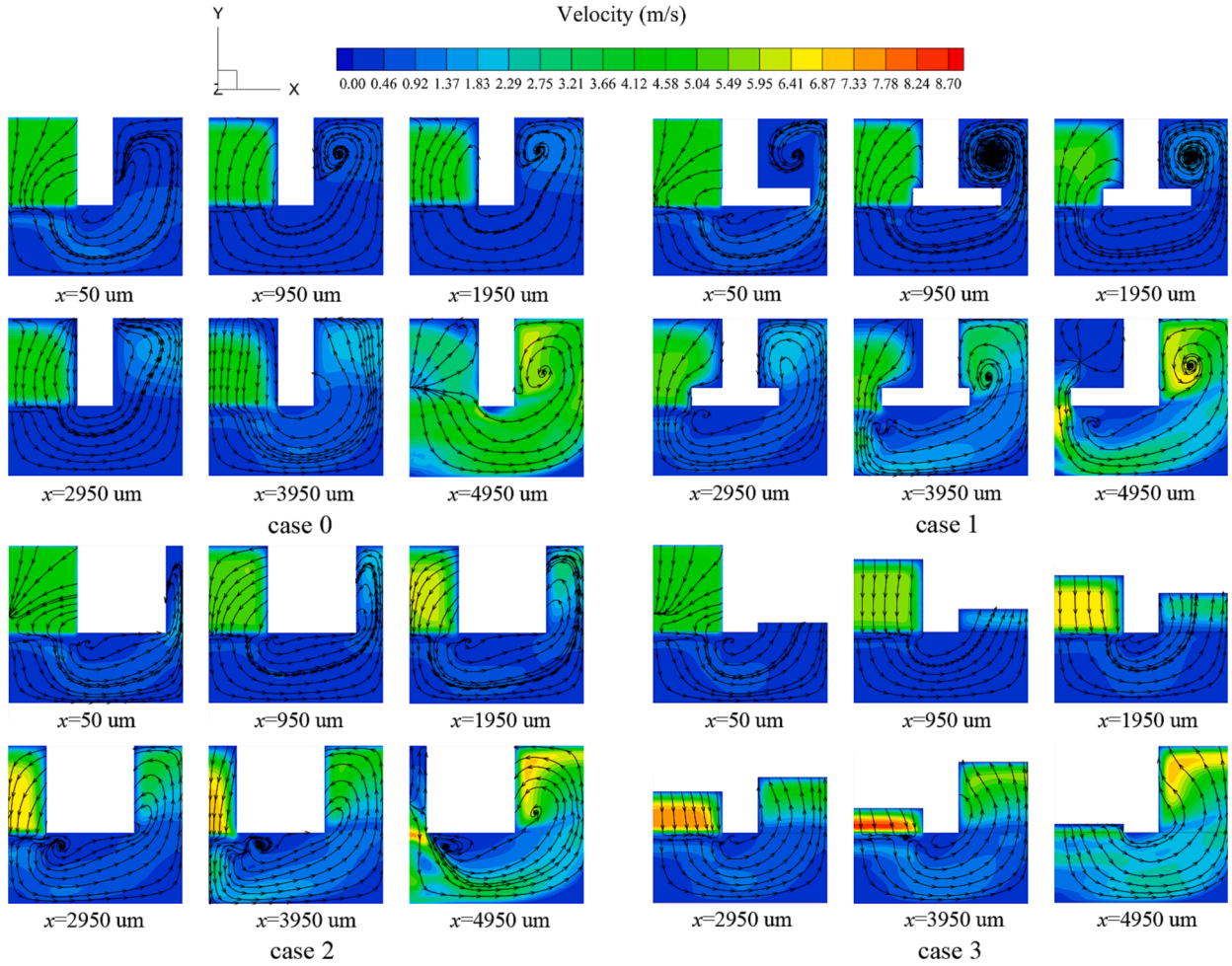


Fig. 11. Velocity distribution of microchannel in y-z planes.

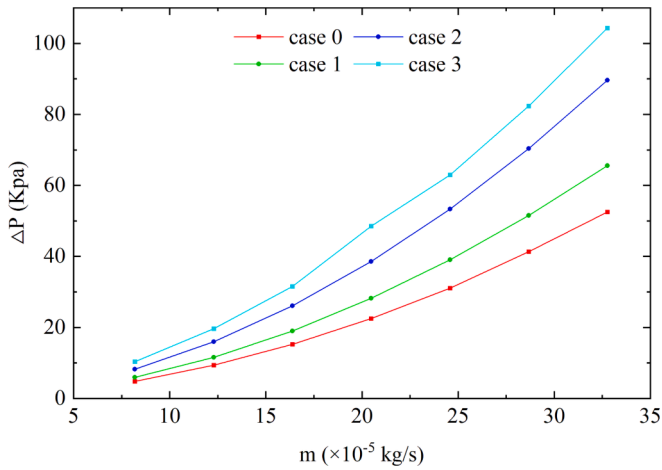


Fig. 12. Pressure drop at different flow rates.

microchannels.

The mean absolute temperature deviation (MATD) $\delta_{T,hs}$ is used to evaluate the non-uniformity of the temperature distribution [34,35], and the expression is as:

$$\delta_{T,hs} = \frac{|T_{base,max} - T_{base,mean}| + |T_{base,min} - T_{base,mean}|}{2} \quad (19)$$

Where the $T_{base,max}$ and $T_{base,min}$ are the maximum temperature and minimum temperature of the heat sink base.

2.3.3. Cooling coefficient of performance

The cooling performance of the heat sink is characterized by cooling coefficient of performance, which is defined as the ratio of heat dissipation to pumping power when the average maximum temperature rise on the surface is 60 K [8]. The relationship is as follows:

$$COP = \frac{Q_{eff,max}}{P} \quad (20)$$

$$P = v\Delta p \quad (21)$$

Where $Q_{eff,max}$ is the heat dissipation when the average maximum temperature rise of the base of the heat sink is 60 K, P is the pumping power, Δp is the pressure drop of the heat sink, the maximum surface average temperature (ΔT_{max}) rise and thermal resistance and heat dissipation have the following relationship:

$$\Delta T_{max} = Q_{eff,max} R_{tot} \quad (22)$$

It can be observed that increasing heat dissipation and decreasing pressure drop are required to enhance COP, and heat dissipation is connected to pressure drop, flow rate, and thermal resistance.

2.4. Mesh independence studies and model validation

The computational domain of the typical MMCHS (case 0) was

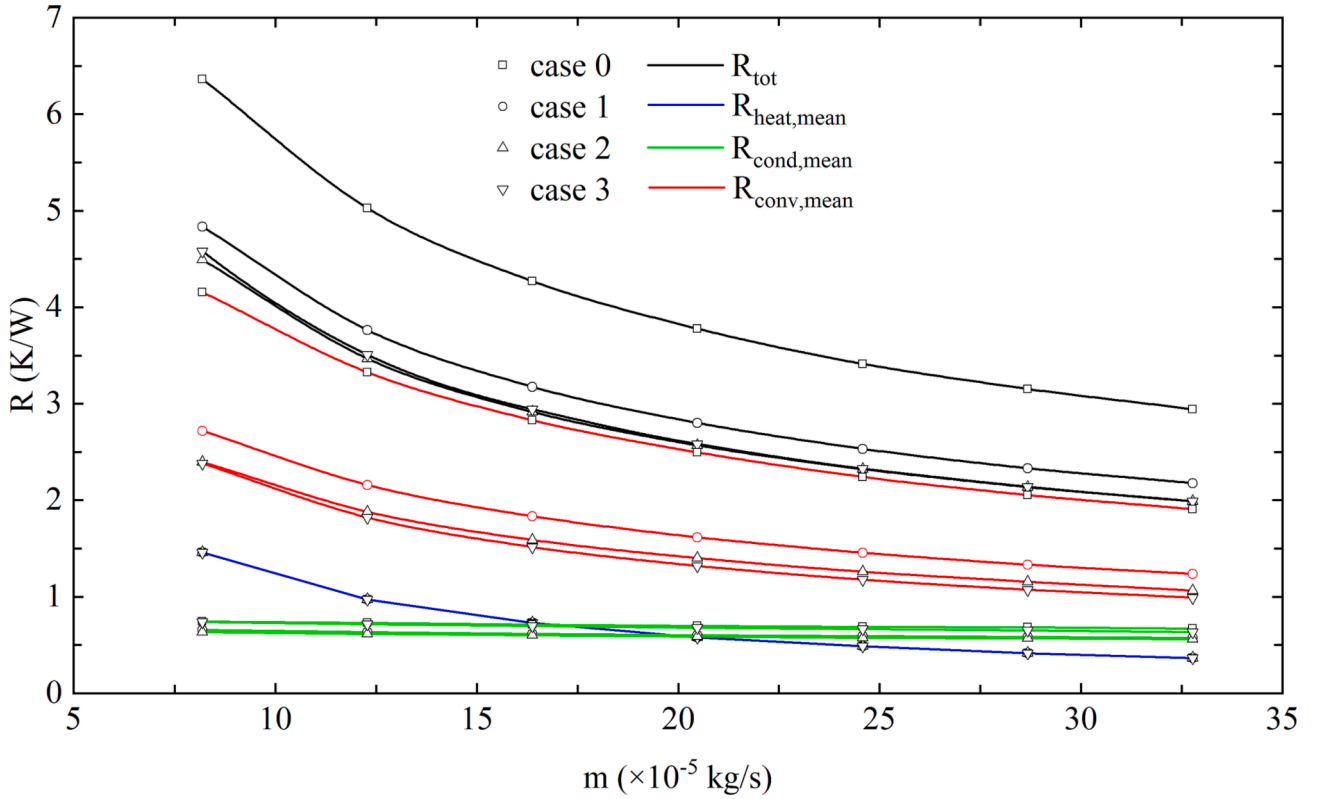


Fig. 13. Detailed contribution of R_{tot} with flow rate change.

meshed using cubic-shaped cells. The regularity of the model resulted in a high-quality meshed grid. The manifold channel inlet, manifold channel outlet, microchannel inlet, microchannel outlet, and bottom grid were refined, and grid independence was verified. The number of elements varied from 232,092 to 10,557,500. Fig. 4 illustrates the independence, characterized by the Δp and the maximum temperature rise on the base surface of the heat sink ($\Delta T_{max,base}$). When the number of grids exceeded 3,150,576, the curves of pressure drop and maximum temperature rise changed smoothly, with fluctuations of less than 1 %. This indicates that the grid density provides sufficient accuracy. To balance simulation accuracy and computational resources, a grid number of 3,150,756 was selected for the simulation. In other cases, which included smaller sharp corners, a combination method of hexahedrons and polyhedrons was used to mesh the computational domain. The grid independence verification and quality are referenced in Table 3.

To verify the accuracy of the numerical calculation, Drummond's experiment [36] was referenced. A numerical simulation was conducted on sample B with a mass flow rate of 1300 kg/m²s, focusing on single-phase heat transfer conditions. The simulation results were then compared to the experimental temperature data obtained from the silicon base surface. As shown in Fig. 5, the simulation results closely align with the experimental data, with a negligible difference within the experimental measurement error range of ± 1 K. This confirms the validity of the present study.

3. Results and discussion

3.1. Non-uniformity of flow distribution

The flow distribution in the microchannel was initially evaluated to facilitate further discussions. Fig. 6 illustrates the distribution of flow in the microchannels when the mass flow rate is 2.048×10^{-4} kg/s, with the microchannels numbered from 1 to 50 along the positive direction of the x-axis. It is evident that the most significant change in flow occurs in

the typical MMCHS microchannel, while the changes in the other three cases are relatively minor. As the microchannel numbering increases, the flow initially experiences a gradual decline, followed by a more pronounced increase.

As depicted in Fig. 7, the average static pressure difference between the inlet and outlet of the microchannel is presented. Static pressure plays a crucial role in the fluid flow within microchannels. It can be observed that the pressure difference between the typical MMCHS and case 3 gradually decreases and then rapidly increases as the number of microchannels increases. This trend aligns with the flow changes observed in the microchannels shown in Fig. 6. However, the pressure difference at the end of case 1 and case 2 decreases. This is attributed to the variation in inlet and outlet flow rates, which arise due to the smaller inlet cross-sectional area of the microchannel compared to the outlet cross-sectional area.

In order to assess the uniformity of flow distribution in the microchannel and its impact on mass flow rate variation, we utilize the flow variance (s^2) as a measure of the uniformity of flow distribution within the microchannel. This parameter represents the deviation of flow rates within the microchannel from the average flow rate. Fig. 8 illustrates that the flow distribution in a typical MMCHS is highly uneven, and as the coolant flow rate increases, the unevenness in flow rate distribution also increases. However, the modified MMCHS exhibits a significantly improved flow distribution. In cases 1 and 2, the unevenness of flow distribution remains relatively unchanged with an increase in coolant flow, while case 3 demonstrates the best uniformity of flow distribution at low coolant flow rates. Comparing the average value of s^2 , the uniformity of flow distribution in the microchannels of case 1, case 2, and case 3 has increased by 70.68 %, 81.44 %, and 83.65 %, respectively, compared to the typical MMCHS. It is important to note that flow distribution uniformity alone does not determine temperature distribution uniformity. The uniformity is also influenced by the distribution of coolant velocity in the microchannels, which leads to varying local thermal resistance of heat sinks. Nevertheless, improving flow

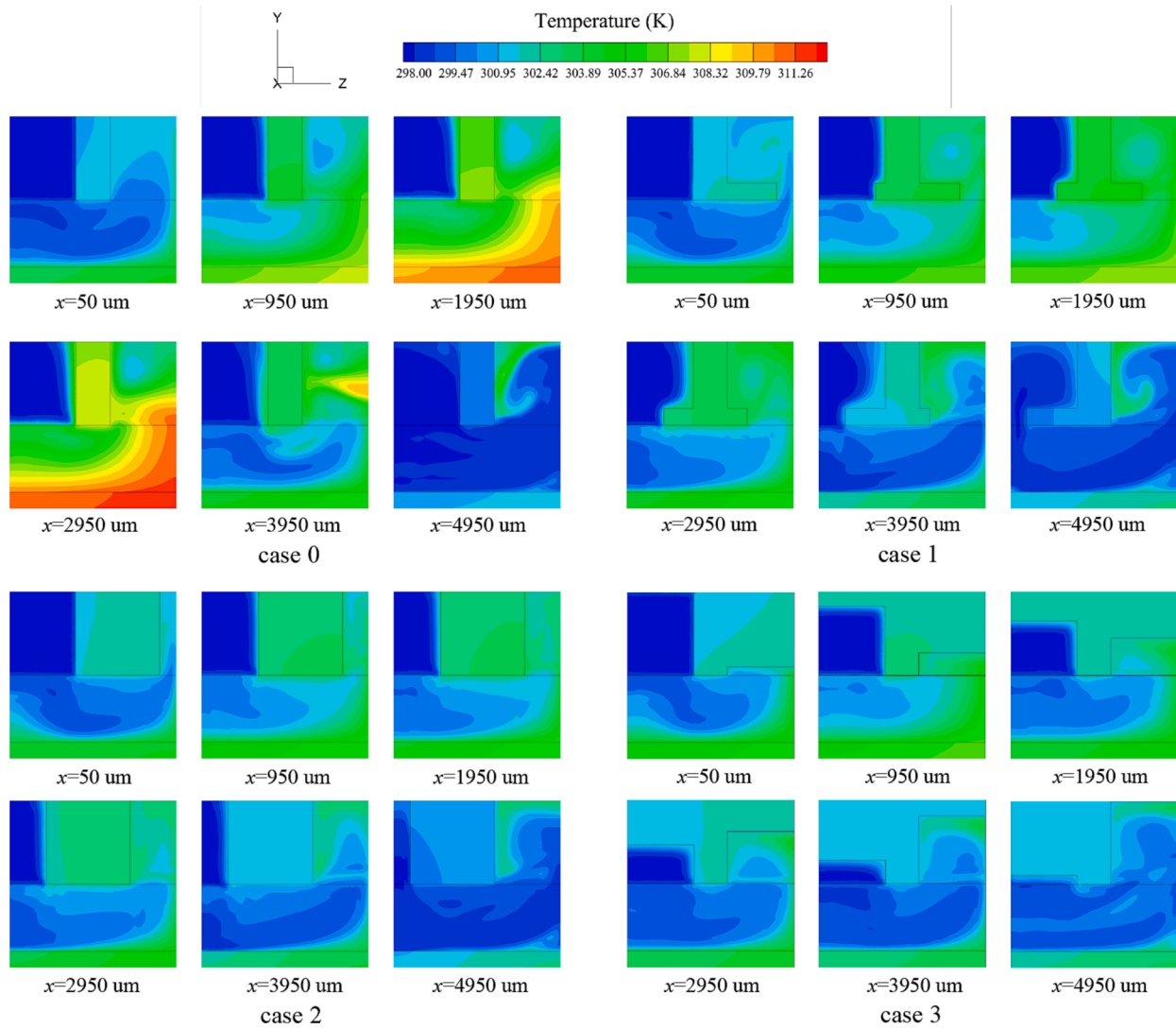


Fig. 14. Temperature distribution of microchannel in y-z plane.

distribution is an effective approach for enhancing temperature distribution uniformity.

3.2. Flow and pressure drop

To observe and explain the flow and heat transfer in the MMCHS, several planes have been established. Fig. 9 illustrates these planes, including 6 y-x planes (orange), 6 y-z planes (green), and a z-x plane (blue). The y-x plane is positioned at the center of the inlet/outlet manifold channel, corresponding to symmetry plane 1 and symmetry plane 2 in Fig. 3. The x-coordinate range for this plane is 0–650 μm , 2250–2750 μm , and 4350–5000 μm . The y-z plane's x-coordinates are 50 μm , 950 μm , 1950 μm , 2950 μm , 3950 μm , and 4950 μm . Lastly, the z-x plane is situated at the base of the heat sink.

Fig. 10 and Fig. 11 illustrate the distribution of fluid velocity on 6 y-x planes and 6 y-z planes, respectively, at a mass flow rate of 2.048×10^{-4} kg/s. In Fig. 10, it can be observed that the fluid in the manifold channel flows perpendicular to the direction of flow into the microchannel. The driving force for the fluid entering the microchannel is primarily static pressure. As the fluid flows from the manifold channel into the microchannel, its flow direction changes. This change in flow direction, along with the action of viscous forces, creates a velocity gradient. Consequently, a local high-speed zone is formed and vortex is generated at the entry of the microchannel, enhancing heat transmission. Near the outlet

of the heat sink in the manifold channel, the fluid flows rapidly into the microchannel due to the obstruction of the wall surface, resulting in excessive flow in the microchannel. Furthermore, Fig. 11 reveals that the fluid velocity is highest in the microchannels near the heat sink outlet ($x = 4950 \mu\text{m}$), while the fluid velocity in the microchannels in the middle of the heat sink ($x = 950 \mu\text{m}$, $x = 1950 \mu\text{m}$, $x = 2950 \mu\text{m}$, $x = 3950 \mu\text{m}$) is considerably lower. This trend aligns with the change in flow rate shown in Fig. 6 for the microchannels. The modified MMCHSs exhibit a significant increase in fluid velocity in the microchannels located in the middle of the heat sink, distinguishing them from typical MMCHSs. Further analysis of the modified MMCHSs will be conducted in the subsequent sections.

For case 1, the addition of a baffle results in a gradual decrease in the cross-sectional area of the microchannel inlet as the x-coordinate increases. Conversely, the cross-sectional area of the microchannel outlet gradually increases. This leads to a gradual increase in the local resistance of the fluid flowing into the microchannel. From Fig. 10, it is evident that the angle between the direction of the fluid velocity at the microchannel outlet in the middle region of the heat sink and the normal of the outlet cross-section decreases when compared to typical MMCHS. As a result, the local resistance at the microchannel outlet decreases, mitigating the decline in flow rate in the microchannels in the middle region of the heat sink. When the x-coordinate is large (near the outlet section of the heat sink), the cross-sectional area of the microchannel

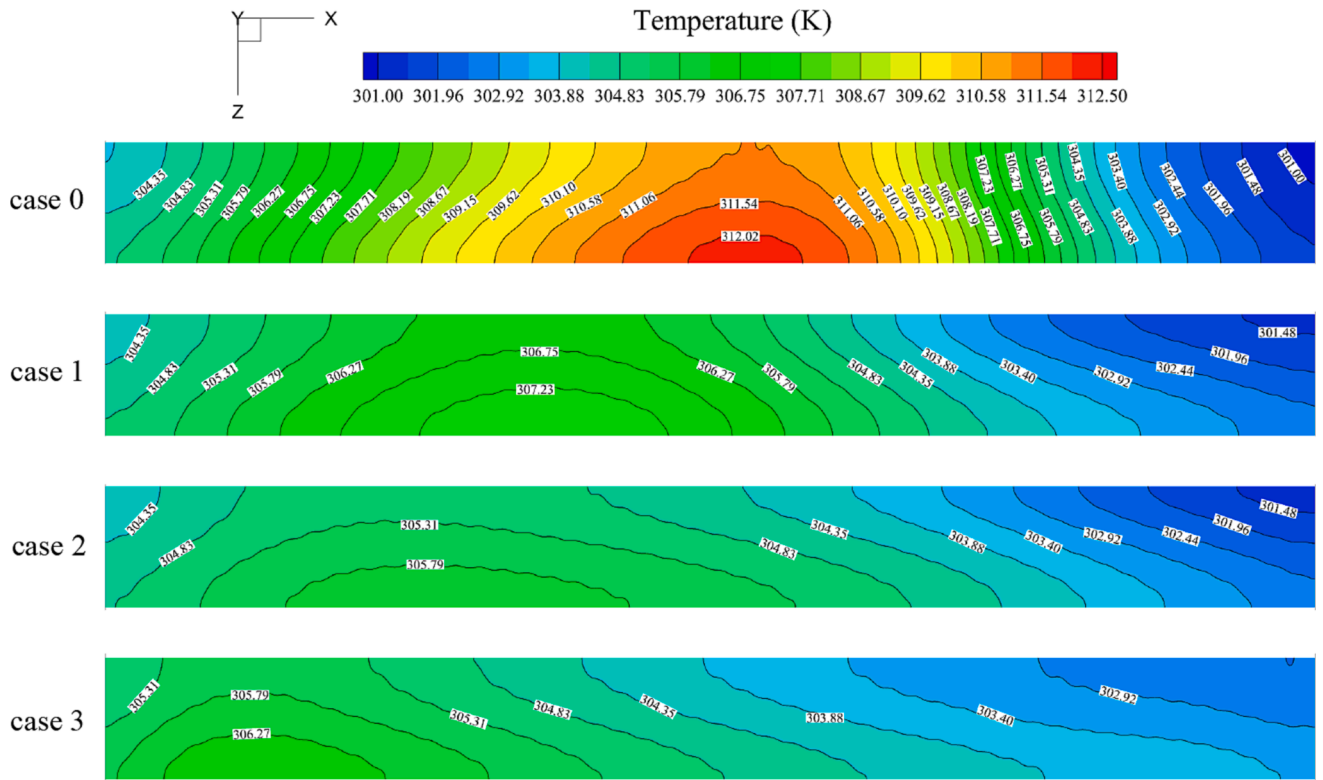


Fig. 15. Uneven temperature distribution at the base of heat sink.

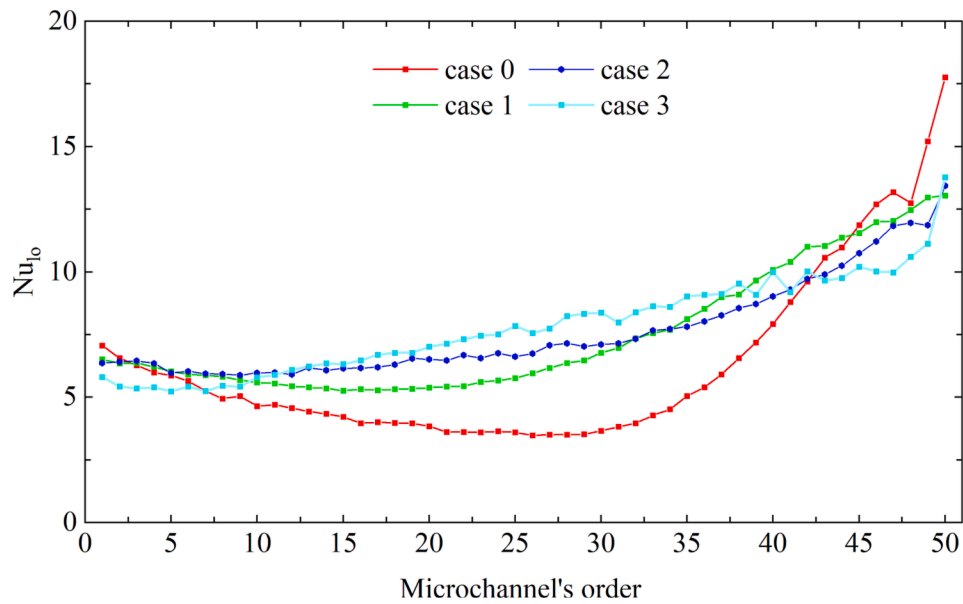


Fig. 16. Local Nusselt number diagram of microchannels.

inlet reaches its minimum, resulting in increased local resistance and improved control of the excessive flow caused by the stagnation pressure. Additionally, the fluid exhibits a larger y-direction component velocity when flowing into the microchannel, creating a high-speed zone that facilitates heat exchange. Simultaneously, a dead zone forms above the end of the inlet manifold channel.

For cases 2 and 3, the flow rate increases within the manifold channel as the x-coordinate increases. This is because the gradual decrease in the inlet manifold channel converts kinetic energy into pressure energy, resulting in increased static pressure and forcing more fluid into the

microchannels in the middle of the heat sink. As a result, there is a more uniform flow distribution. However, there are some differences between case 2 and case 3. In case 3, only the size of the manifold channel changes while the size of the microchannel inlet and outlet remains the same. On the other hand, in case 2, both the size of the microchannel inlet and outlet, as well as the size of the manifold channel, change. By combining Fig. 10 and Fig. 11, we can observe that in case 3, the fluid is squeezed by the upper wall of the manifold channel and the 'mainstream area', resulting in a combined velocity inclined towards the negative direction of the y-axis. This leads to a smaller angle at which the flow

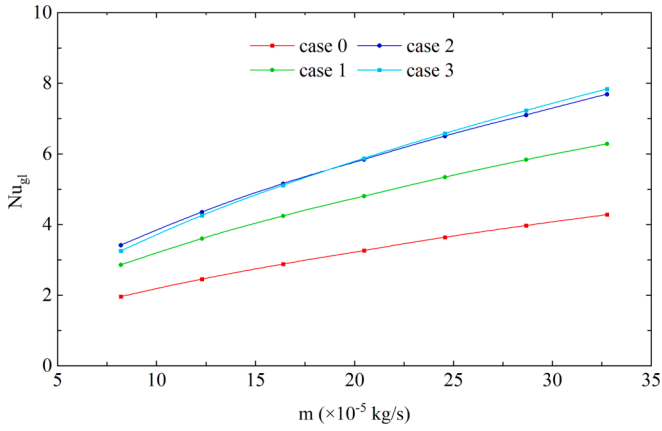


Fig. 17. The relationship global Nusselt number and flow rate.

velocity changes when entering the microchannel from the manifold channel, thereby improving the uniformity of flow distribution. However, it also increases the flow resistance in the manifold channel. In case 2, as the cross-section of the manifold channel decreases with increasing x-coordinate, the fluid is squeezed by the side walls of the manifold channel. As shown in Fig. 11, when the x-coordinate is small, the fluid flows from the manifold channel into the microchannel at a greater flow rate compared to case 0 and case 1. This results in a more obvious vortex generated by the fluid entering the microchannel. Additionally, as the x-coordinate increases, the size of the microchannel inlet decreases. While this improves flow distribution, it also increases the flow resistance in the microchannel section. Therefore, as depicted in Fig. 7, case 2 has the largest pressure difference for most microchannels. Fig. 12 illustrates the relationship between pressure drop and flow rate. It is evident that case 3 experiences the highest pressure loss compared to the other three cases. Specifically, case 1, case 2, and case 3 exhibit an average increase in pressure drop of 24.65 %, 70.99 %, and 106.96 %, respectively, when compared to a typical MMCHS.

3.3. Convective heat transfer and temperature distribution

Fig. 13, Fig. 14, and Fig. 15 illustrate the changes in total average thermal resistance and the thermal resistance of each component with varying flow rates. They also depict the temperature distribution on the

y-z plane and at the base of the heat sink for a mass flow rate of 2.048×10^{-4} kg/s. From Fig. 13, it is evident that a typical MMCHS exhibits the highest total thermal resistance, while the modified MMCHSs show significant reductions in thermal resistance. The total thermal resistance of case 1, case 2, and case 3 decreased by an average of 25.48 %, 31.52 %, and 31.03 % respectively, when compared to a typical MMCHS. The convective heat transfer resistance is the primary factor influencing changes in total thermal resistance, as the thermal conduction resistance remains relatively unchanged regardless of manifold arrangement and the addition of baffles. This is due to the decisive role played by the dimensions of microchannels and fins in thermal conduction resistance, with the microchannel arrangement being consistent across all four cases. However, it is observed that the rate of decrease in thermal resistance slows down as the flow rate increases. This suggests that beyond a certain point, increasing the flow rate to reduce thermal resistance becomes less cost-effective, and structural improvements are required. Furthermore, the local thermal resistance significantly impacts the uniformity of temperature distribution at the base of the heat sink. Fig. 16 illustrates the changes in local Nu in the microchannels. When combined with Fig. 6, it can be observed that the variations in local Nu align with the changes in flow rate. In typical MMCHS, the microchannels near the heat sink outlet area experience the highest flow rate

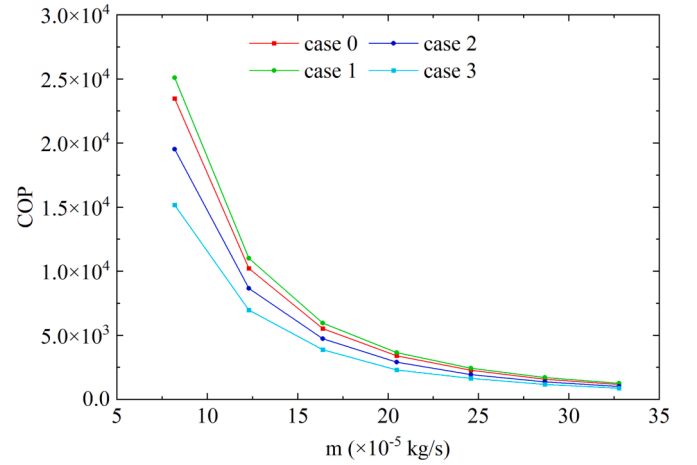


Fig. 19. The COP variations at different flow rates.

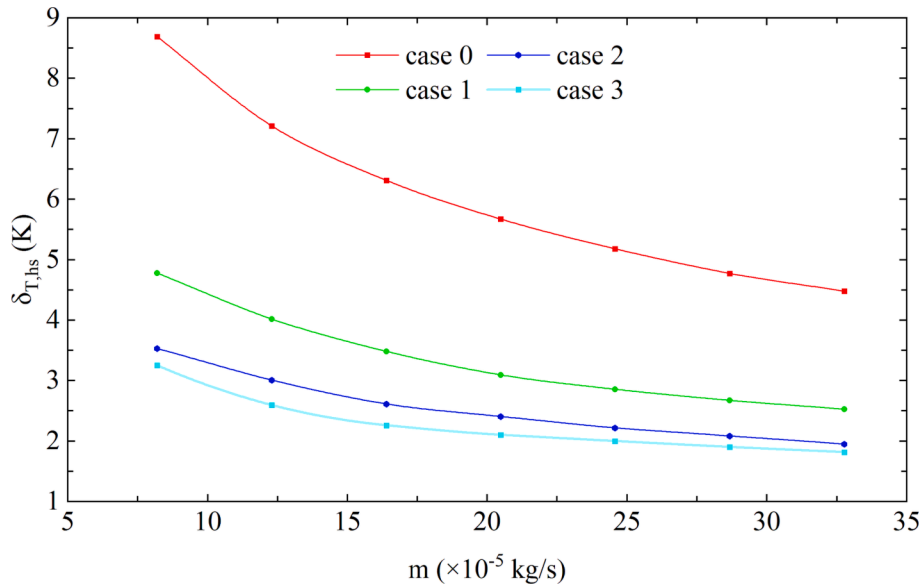


Fig. 18. Relationship between temperature non-uniformity and flow rate.

and velocity, while the microchannels in the middle area of the heat sink have the lowest flow rate and velocity. Consequently, the Nu in the middle area of the heat sink is smaller, resulting in larger thermal resistance and localized high temperature, as depicted in Fig. 15. However, in other cases, the enhancement of flow distribution uniformity leads to a smoother local Nu curve and a significant improvement in temperature distribution uniformity at the heat sink base. The modification of the heat sink sacrifices heat exchange intensity near the outlet but reduces local thermal resistance in the middle region, thereby enhancing the uniformity of temperature distribution at the base of the heat sink.

As depicted in Fig. 14 and Fig. 15, the temperature distribution difference in typical MMCHS is also the largest. The change in fluid flow rate in microchannels has a significant impact on heat exchange, as illustrated in Fig. 6. As the x-coordinate increases, the fluid flow rate in microchannels initially decreases slightly and then increases significantly. Consequently, this results in a smaller temperature rise in the microchannels when the x-coordinate is large. At $x = 4950 \text{ } \mu\text{m}$, for a typical MMCHS, the temperature rise of the fluid is minimal. In other words, a substantial amount of fluid flows out of the microchannel without being fully heated, leading to a waste of cooling capacity. When examining Fig. 11 in conjunction with this, it becomes evident that areas with high flow rates typically exhibit lower temperatures. However, an abnormal phenomenon is observed in case 1, case 2, and case 3, as shown in Fig. 14, at $x = 4950 \text{ } \mu\text{m}$. In these instances, the temperature in areas with higher fluid flow rates in microchannels is higher than that near the bottom wall surface. This can be attributed to the influence of stronger vortices, which enhance heat exchange between the fluid and the side wall surface, consequently resulting in a temperature increase. The density of isotherms provides an indication of the magnitude of the temperature gradient on the plane. Based on the analysis of Fig. 15, it is evident that the temperature in the high-temperature region of modified MMCHSs has noticeably decreased along with the temperature gradient. Notably, case 2 and case 3 exhibit the most sparse isotherms and the most uniform temperature distribution. This can be attributed to the higher flow rate in the middle area microchannels of the heat sink in these cases compared to a typical MMCHS. Consequently, the fluid enters the microchannels at a higher velocity, resulting in a stronger jet impact and improved heat exchange in the middle region of the heat sink, thereby enhancing temperature distribution uniformity. However, in case 1, case 2, and case 3, where the flow rate distribution is similar near the inlet section of the heat sink, the enhancement effect of heat exchange in the modified MMCHS is comparatively weaker. Consequently, the local high-temperature areas gradually shift towards the negative x direction.

Fig. 17 and Fig. 18 illustrate the relationship between the global Nu of the heat sink, the uniformity of the temperature distribution at the base, and the flow rate. It is observed from Fig. 17 that the global Nu increases as the flow rate increases, which is contrary to the trend of convective heat transfer thermal resistance shown in Fig. 13. In comparison to the typical MMCHS, the average global Nu of case 1, case 2, and case 3 increased by 31.90 %, 43.98 %, and 44.06 % respectively. Fig. 18 demonstrates that the temperature distribution uniformity of the modified MMCHS has significantly improved. Moreover, as the flow rate increases, the temperature uniformity further improves, although the rate of improvement slows down. The average temperature distribution uniformity of case 1, case 2, and case 3 is 44.55 %, 57.71 %, and 62.13 % higher respectively than that of the typical MMCHS.

3.4. Comprehensive assessment of thermal performance

Through the previous analysis, it was found that the flow distribution and Nu of the modified MMCHS have been significantly improved. However, it was also observed that the pressure drop has increased. Additionally, it appears that as the flow rate increases, the rate of improvement in flow distribution and Nu slows down, while the increase

in pressure drop continues to rise. It is important to note that a single indicator is insufficient to measure the comprehensive cooling efficiency. In order to evaluate the heat dissipation performance of a heat sink, the COP is used, which represents the ratio of heat dissipation to pump power. A higher COP indicates that the heat sink consumes less energy to dissipate the same amount of heat. Fig. 19 illustrates the relationship between COP and mass flow rate. Unanimously, as the mass flow rate increases, the COP decreases for all cases. This suggests that when designing a heat sink, the inlet flow rate can be appropriately reduced to save energy while still meeting the heat dissipation capacity. The increase in pressure drop in case 2 and case 3 has a greater impact on the COP compared to the decrease in thermal resistance. Consequently, the average COP values of case 2 and case 3 decreased by 14.58 % and 29.81 % respectively, when compared to the typical MMCHS. Although the heat dissipation uniformity of case 2 and case 3 has improved, their overall heat dissipation performance has decreased. On the other hand, case 1 takes into account both heat dissipation uniformity and heat dissipation performance. The average COP value increased by 7.67 % compared to the typical MMCHS, indicating that adding flow-regulating baffles between the manifold channels and microchannels is a favorable design.

4. Conclusion

In this study, a new modified MMCHS was proposed and an extended investigation of the MMCHS for divergent/convergent manifold channels was conducted. The study examined the flow distribution, pressure drop, velocity distribution, temperature uniformity, Nusselt number, thermal resistance, and comprehensive heat exchange performance of the heat sink under various flow rates and uniform heat flux using numerical simulation. The following conclusions were drawn:

- (1) The addition of flow-regulating baffles between the manifold channel and microchannel in the MMCHS resulted in the best cooling performance. It also significantly improved the temperature distribution uniformity at the base of the heat sink, thereby achieving a balance between cooling performance and heat dissipation uniformity. Compared to the typical MMCHS, the COP value increased by 7.67 %, the average microchannel flow distribution uniformity increased by 70.68 %, and the temperature distribution uniformity at the base of the heat sink increased by 44.55 %.
- (2) The MMCHS, which utilizes divergent/convergent manifold channels, exhibits the most uniform heat dissipation. However, its cooling performance has declined. Specifically, the temperature distribution uniformity at the base of the MMCHS diverging/converging in the z direction has increased by 57.71 %, while the uniformity of flow distribution in the microchannels has increased by 81.44 %. Additionally, the COP has decreased by 14.58 %. Similarly, the temperature distribution uniformity of the MMCHS diverging/converging in the y direction has increased by 62.13 %, with the uniformity of flow distribution experiencing an 83.65 % increase. However, the COP has decreased by 29.81 %.
- (3) The fluid flow rate has a significant impact on thermal resistance and global Nu. In the modified MMCHS, the main change in thermal resistance is due to convective heat transfer. The average total thermal resistance of the MMCHS in the diverging/converging manifold channels in the z and y directions decreased by 31.52 % and 31.03 % respectively, while the global Nu increased by an average of 43.98 % and 44.06 %. Additionally, by adding a baffle, the total thermal resistance of the MMCHS decreased by an average of 25.48 %, and the global Nu increased by an average of 31.90 %.
- (4) The average pressure drop of MMCHS, when using z and y direction divergent/convergent manifold channels, increased by

70.99 % and 106.96 % respectively, compared to typical MMCHS. This significant increase in pressure drop has a greater impact on the COP than the thermal resistance on COP. However, when adding a baffle, the average pressure drop of MMCHS only increased by 24.65 %.

In summary, the modified MMCHS demonstrates satisfactory temperature distribution uniformity and effective heat dissipation performance. However, it is important to note that there are several parameters that influence the heat transfer performance and uniformity of heat sinks. Therefore, further investigation can be conducted to explore additional aspects mentioned in this survey, such as examining the size of the baffle, the angle of the diverging/converging channel, and considering the use of curved diverging/converging paths instead of straight lines, among others.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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